

SANTA MARIA BUOY 46011, CA, USA

WMO: 992960

Lat: 34.956N

Lon: 121.019W

Elev: 0

StdP: 101.33

Time Zone: -8.00 (NAP)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	8.7	9.3	N/A	N/A	N/A	N/A	N/A	N/A	15.5	11.3	14.5	11.0	5.2	330	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	1.9	19.0	N/A	18.3	N/A	17.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.6	11.7	10.8	7.3	20.5	1.3	1.9	6.4	21.9	5.6	23.0	4.9	24.1	4.0	25.5		
			DB	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	12.5	12.1	12.0	11.6	11.9	12.8	14.0	14.6	14.9	15.0	14.2	12.8
	DBStd	1.86	1.70	1.48	1.37	1.13	0.97	1.02	1.46	1.38	1.55	1.67	1.49	1.48
	HDD10.0	4	2	1	1	1	0	0	0	0	0	0	0	0
	HDD18.3	1875	181	174	197	203	199	167	135	116	103	104	124	171
	CDD10.0	1173	79	61	62	48	59	83	123	142	147	155	126	87
	CDD18.3	1	0	0	0	0	0	0	0	0	1	1	0	0
	CDH23.3	1	0	0	0	1	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.7	5.3	5.9	6.1	6.6	6.6	6.2	5.0	5.2	5.1	5.1	5.4	5.5
Precipitation	PrecAvg	356	70	75	54	27	14	4	3	2	4	17	30	55
	PrecMax	722	310	328	247	97	60	18	13	9	17	83	99	184
	PrecMin	116	5	4	1	6	3	0	0	0	0	2	5	5
	PrecStd	141	67	67	53	24	13	4	3	2	3	18	23	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	16.7	16.8	16.0	16.1	16.5	19.1	18.6	20.0	19.8	18.4	17.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	16.1	15.7	15.5	14.1	14.5	15.3	17.8	18.0	19.0	19.0	17.4	16.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	15.4	15.0	14.6	13.5	13.8	14.8	16.9	17.5	18.0	18.4	16.7	15.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	14.8	14.1	13.9	13.0	13.4	14.2	16.2	16.9	17.2	17.7	16.2	15.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.3	2.0	1.9	1.7	1.5	1.6	1.6	1.7	1.8	1.9	2.2	2.2
		MCDBR	3.0	2.7	2.6	2.5	2.1	2.2	1.8	1.7	1.9	2.3	2.8	2.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.313	0.324	0.339	0.347	0.354	0.352	0.357	0.344	0.347	0.346	0.321	0.319	
	taud	2.587	2.566	2.520	2.491	2.493	2.491	2.533	2.546	2.533	2.538	2.573	2.580	
	Ebn at Noon	899	926	935	938	930	926	919	929	913	887	881	869	
	Edh at Noon	77	87	99	107	108	108	103	100	97	89	77	74	
All-Sky Solar Radiation	RadAvg	2.93	3.78	5.13	6.37	7.00	7.17	6.46	5.97	5.41	4.25	3.27	2.60	
	RadStd	0.23	0.39	0.35	0.44	0.52	0.40	0.35	0.40	0.36	0.33	0.20	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.53	N/A	-1.28	+0.73	N/A	N/A	N/A	-165	+204	+44

Nomenclature: See separate page